Algorithms

TREES & UNDIRECTED DFS

Last Week

- Euler Paths & Cycles
- De Bruijn Sequence
- Hamiltonian Paths & Cycles

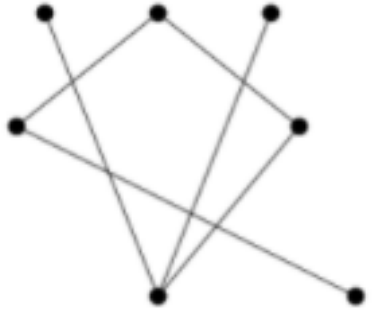
This Week

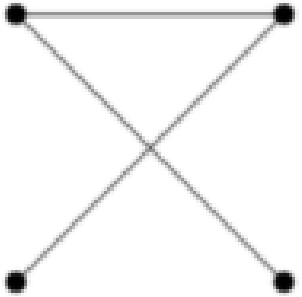
- Trees
- Undirected DFS

Trees – Undirected Graphs

- A tree is an undirected graph $G = (V, E)$ that satisfies any of the following conditions:
  - G is acyclic and connected.
  - G is acyclic and adding any edge will form a cycle.
  - G is connected and deleting any edge will disconnect G.
  - Every two vertices have a unique path connecting them.

Identify The trees:





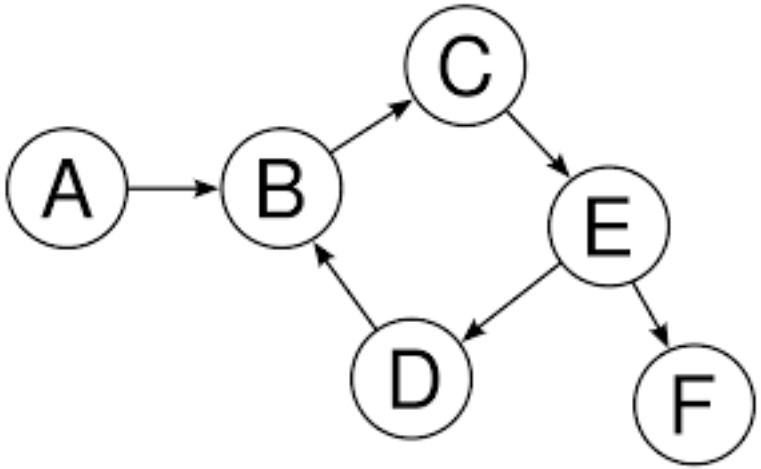


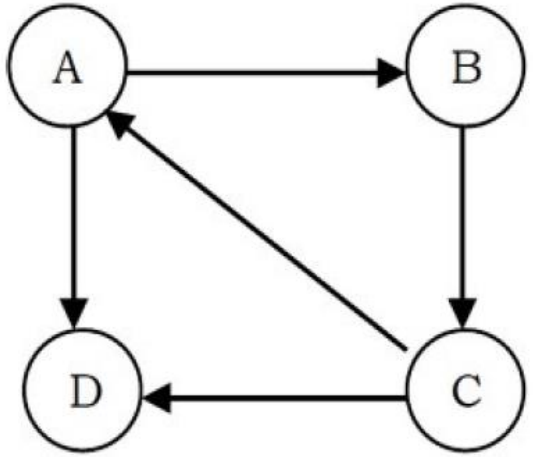


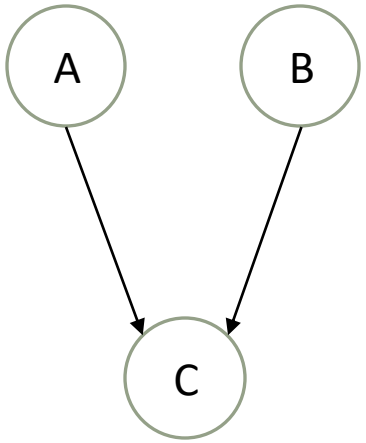
Trees – Undirected Graphs

Root In a Directed Graph

- $r \in V$ is a root of a directed graph $G = (V, E)$ if for every vertex $v \in V$ there is a directed path from r to v.
- **Not only for trees!**
- Identify the roots:

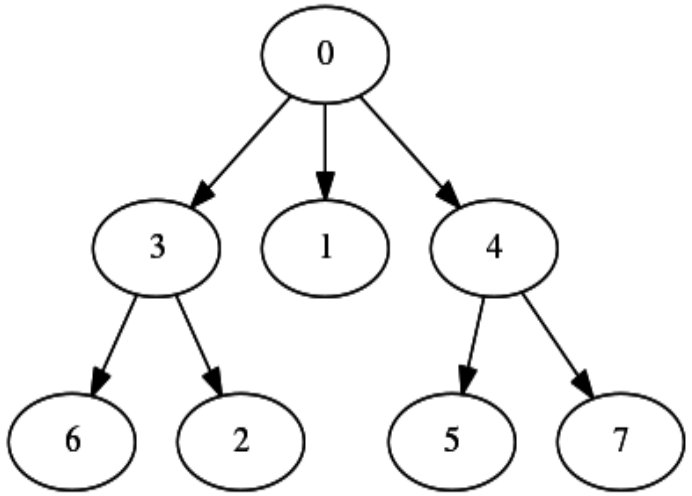






Directed Tree

- A directed tree is a directed graph that has a root and its underlying undirected graph is a tree.



Exercise 1

- **Definition:** Let T be an undirected tree. A leaf in T is a node whose degree is 1.
- **Lemma 1:**
  - Every undirected tree over $n \geq 2$ nodes has at least two leaves.
  - We will prove the lemma in two different ways:
    - by counting arguments
    - by induction

Proof 1 – Counting Arguments

- Let T be an undirected tree and let x be the number of leaves in T
- The number of non-leaf vertices in T is $n - x$
- Reminder: handshake theorem:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Proof 1 – Counting Arguments

- Using the handshake theorem we obtain:

$$2m = \sum_{v \in V} \deg(v)$$

$$= \sum_{v \text{ is a leaf}} \deg(v) + \sum_{v \text{ is not a leaf}} \deg(v)$$

$$\geq x \cdot 1 + (n - x) \cdot 2$$

$$= 2n - x$$

Proof 1 – Counting Arguments

- We saw in class that the number of edges in T is $m = n - 1.$
- We proved that $2m \geq 2n - x$
- Therefore, $2n - 2 \geq 2n - x$
- $\Longrightarrow x \geq 2$



Proof 2 – Induction

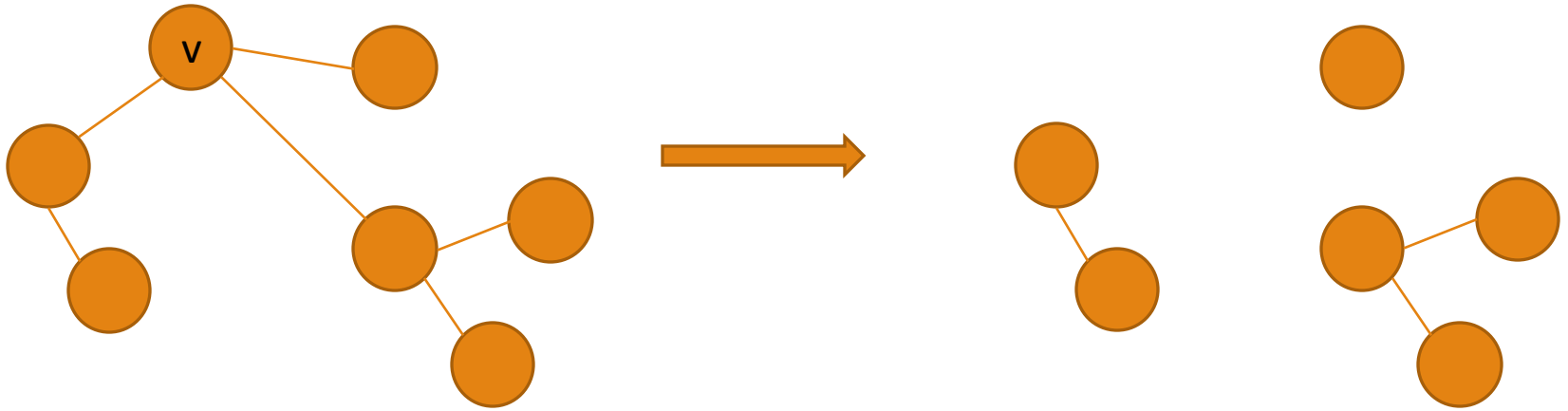
- We prove using induction over n
- Base case: A tree of size $n = 2$ has 2 leaves.
- Inductive hypothesis:
  - Every tree of size between 2 and $n - 1$ has at least 2 leaves.



Proof 2 – Induction

- Inductive step:
  - Let T be an undirected tree of size n and let v be any vertex in T. Let $\deg(v) = k$.
  - Remove v and its adjacent edges from T to obtain k sub-trees $T_1, \ldots, T_k$, each of them over less than n nodes.

$Example$:
$k = 3$

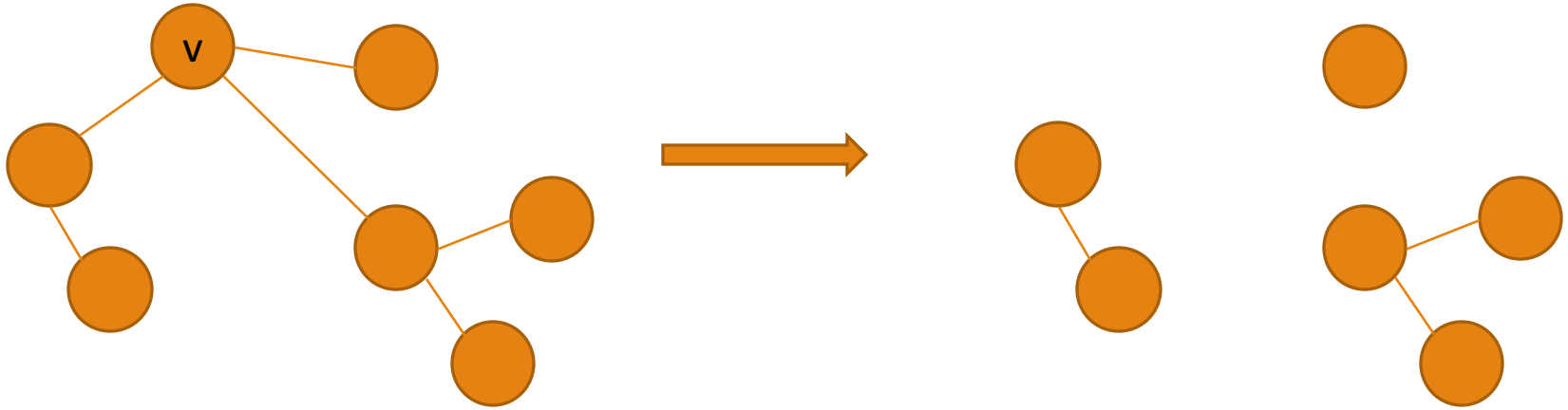


Proof 2 – Induction

- By the induction hypothesis, if $|T_i| \geq 2$ then it has at least two leaves.
- At most one of them is adjacent to v in T, therefore T_i includes at least one leaf of T.
- If $|T_i| = 1$ then the single node of T_i is a leaf in T.

Proof 2 – Induction

- Therefore, every T_i includes at least one leaf of T and T has at least k leaves.
- If $k = 1$ then v itself is a leaf, and T_i includes at least one leaf.



DFS - Depth First Search

A depth-first search algorithm visits the child vertices first. It traverses the depth of any particular path before exploring its breadth.

Algorithm 1 DFS(G, v)

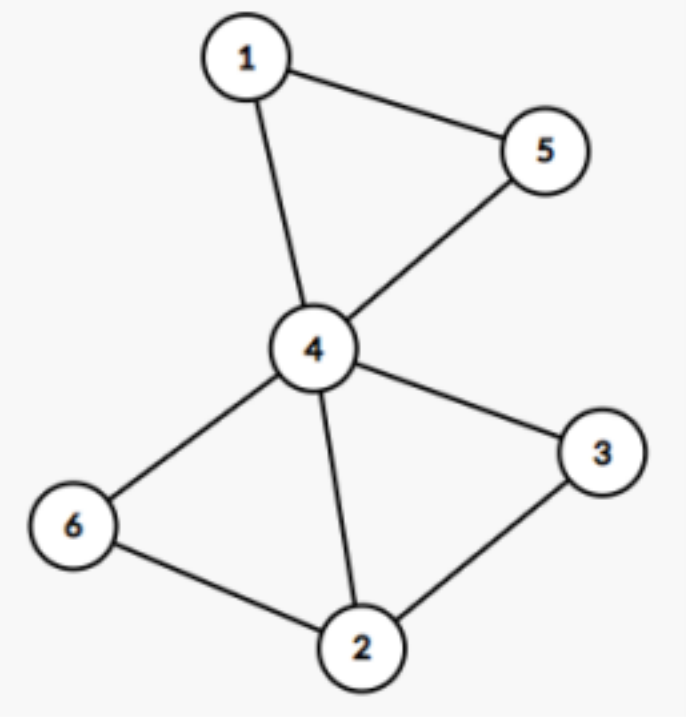
```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

$DFS(4)$



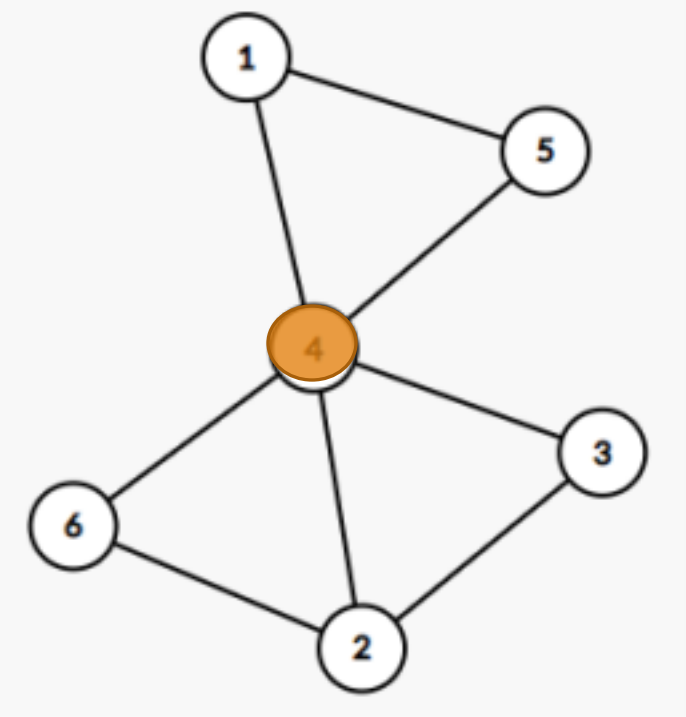
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(4)$

$DFS(4)$



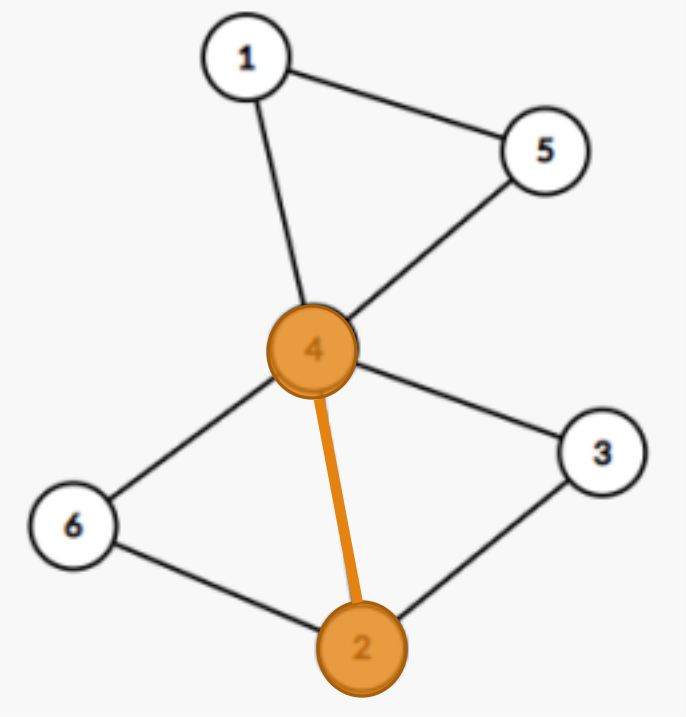
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(2)$
$DFS(4)$

$DFS(4)$



DFS - Example

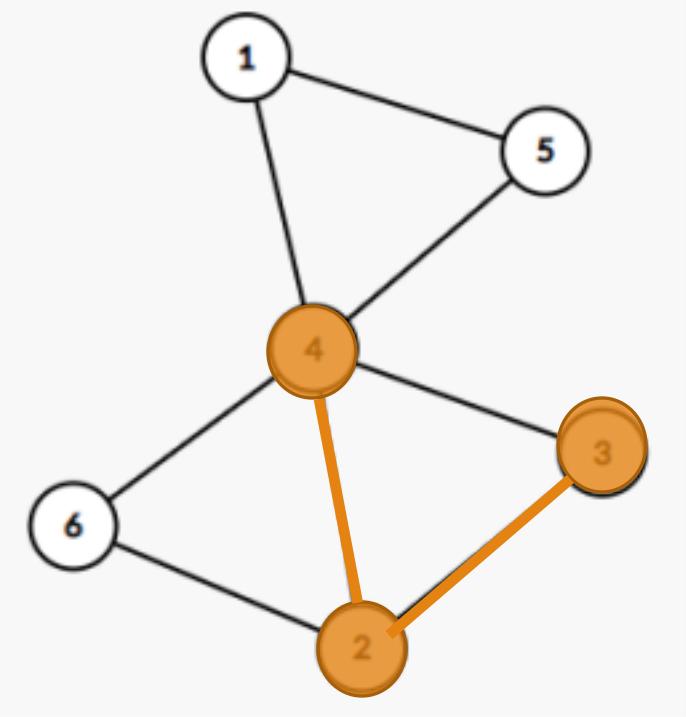
Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack:

$DFS(3)$
$DFS(2)$
$DFS(4)$

$DFS(4)$



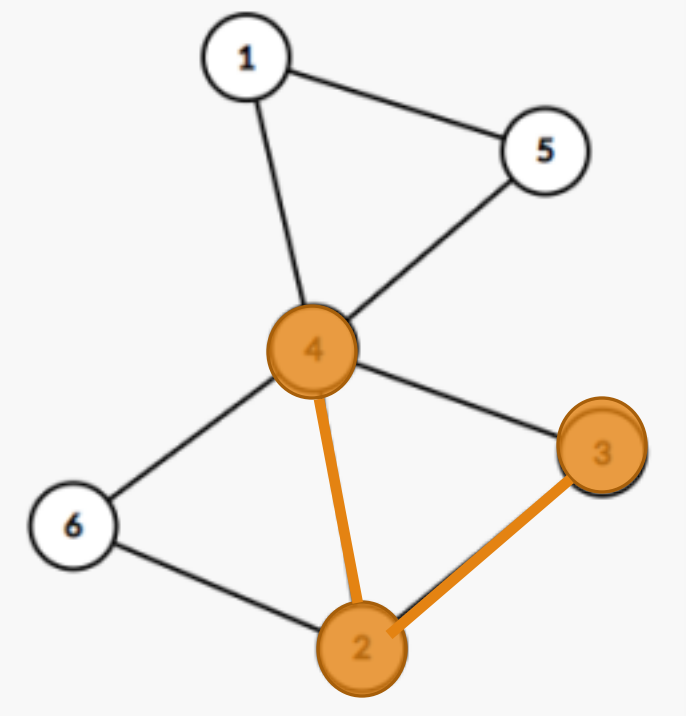
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(2)$
$DFS(4)$

$DFS(4)$

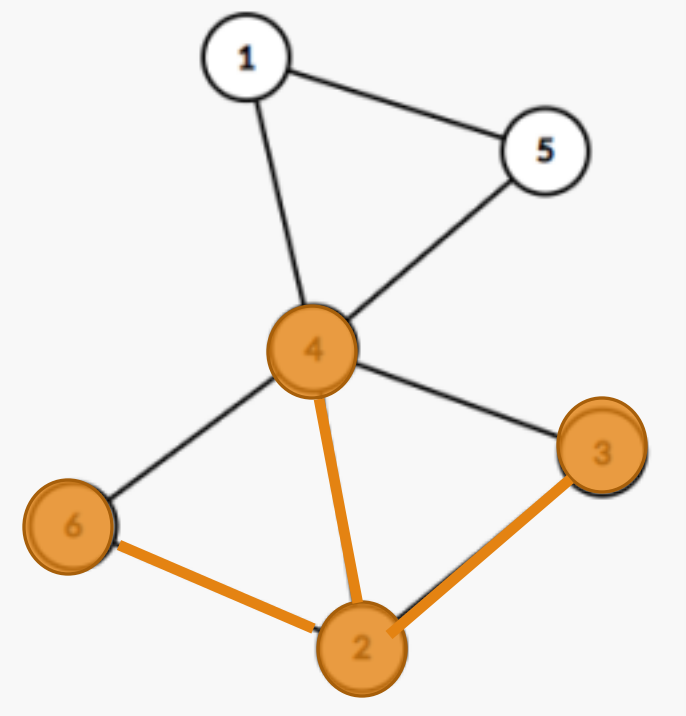


DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

$DFS(4)$



Running Stack:

$DFS(6)$
$DFS(2)$
$DFS(4)$

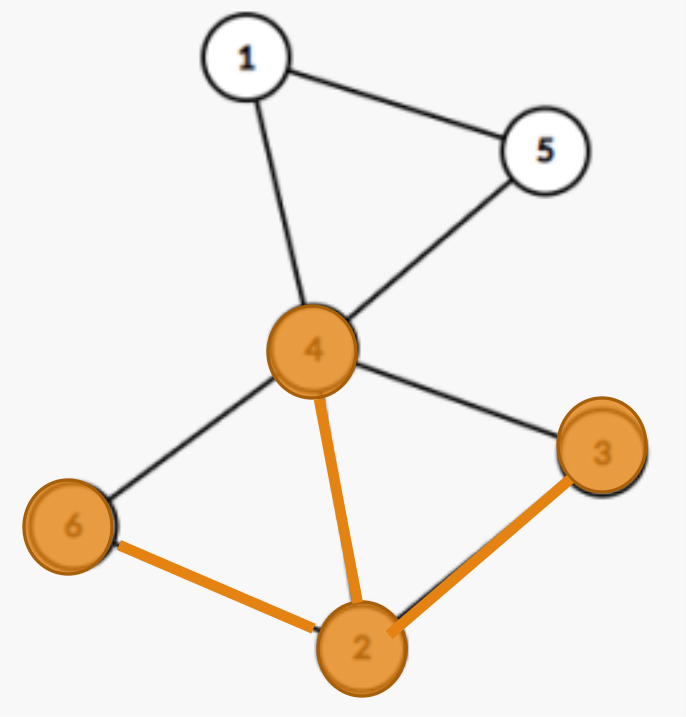
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(2)$
$DFS(4)$

$DFS(4)$



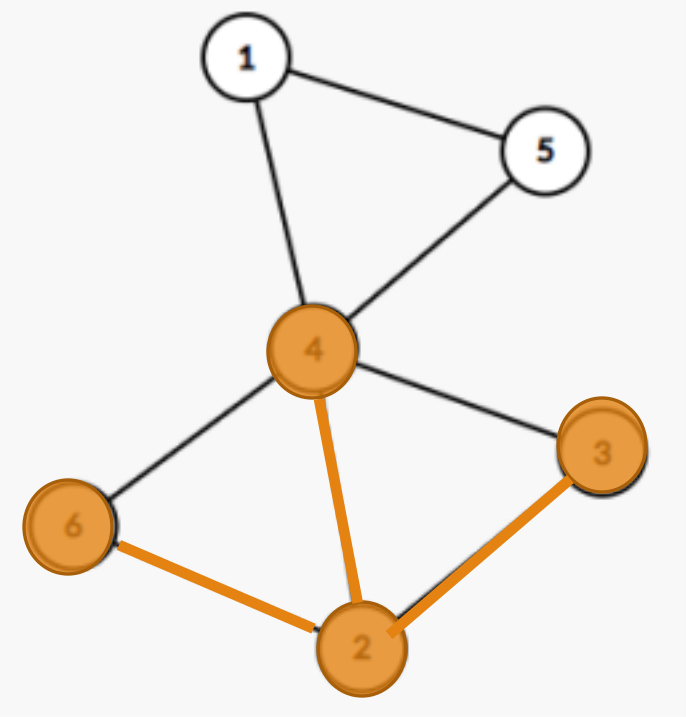
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(4)$

$DFS(4)$



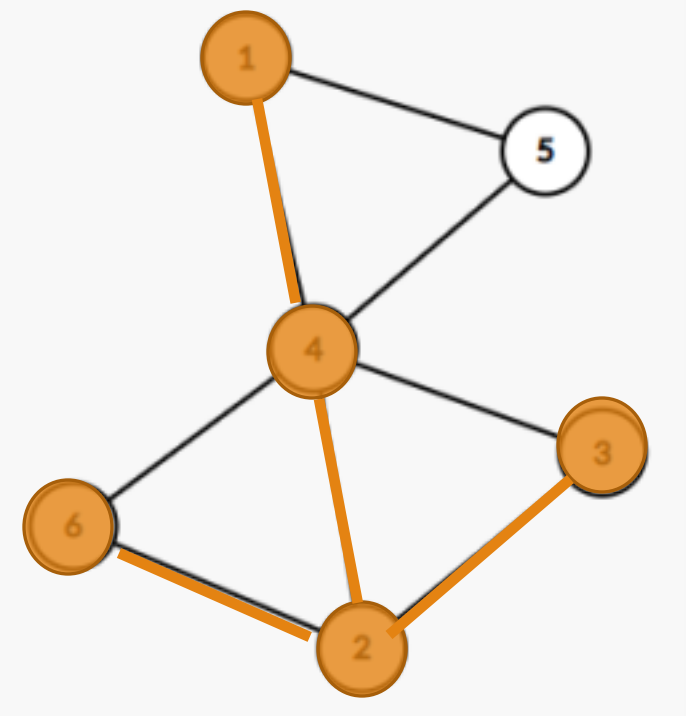
DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

Running Stack: $DFS(1)$
$DFS(4)$

$DFS(4)$

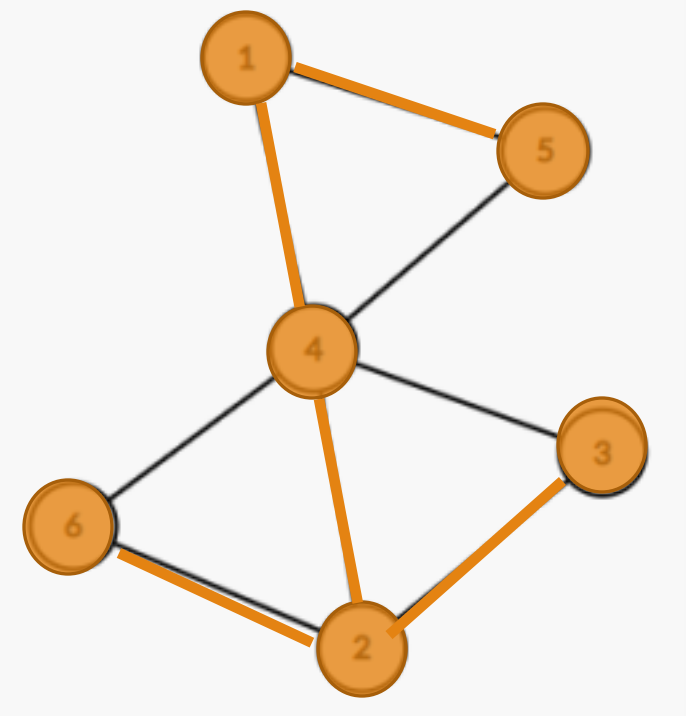


DFS - Example

Algorithm 1 DFS(G, v)

```
1: mark v as explored
2: for each u adjacent to v do
3:     if u is unmarked then
4:         DFS(G, u)
```

$DFS(4)$

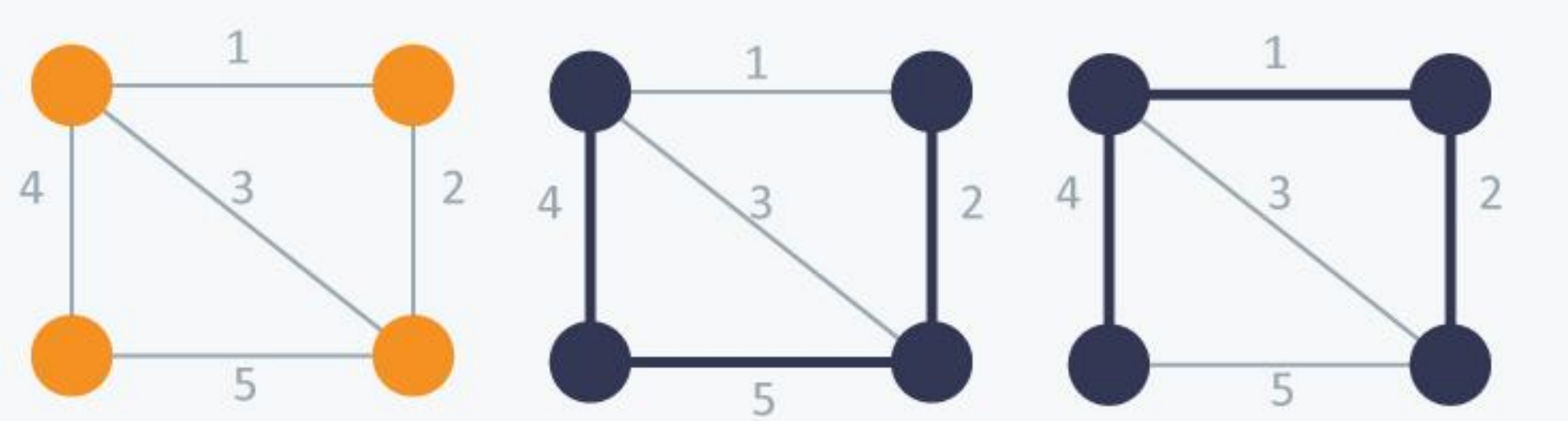


Running Stack:

$DFS(5)$
$DFS(1)$
$DFS(4)$

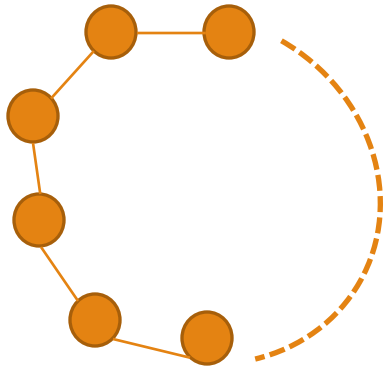
Spanning Tree

$T = (V', E')$ is a spanning tree of $G = (V, E)$ if it is a tree and $V' = V, E' \subseteq E$.

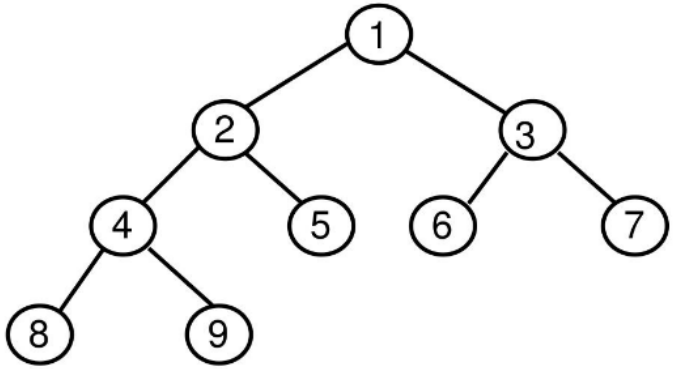


Spanning Tree - Exercise

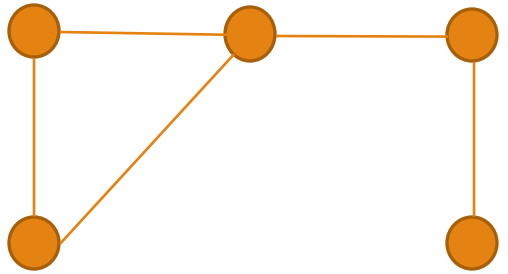
How many spanning trees the following graphs have?



Cycle with n nodes

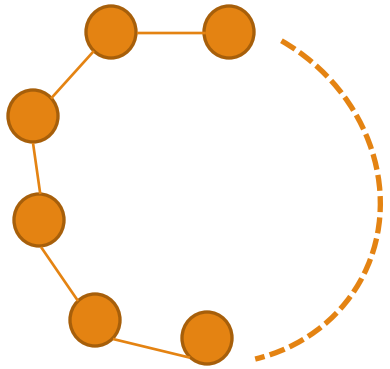


Tree with 9 nodes



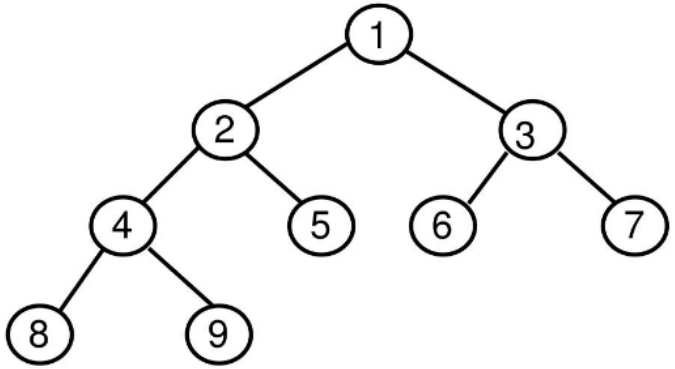
Spanning Tree - Solution

How many spanning trees the following graphs has?



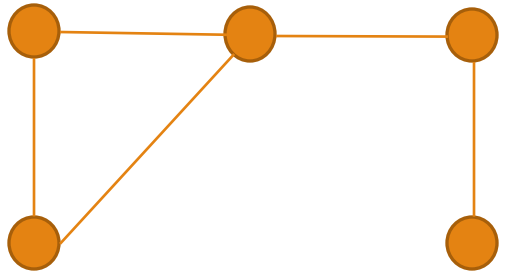
Cycle with n nodes

n



Tree with 9 nodes

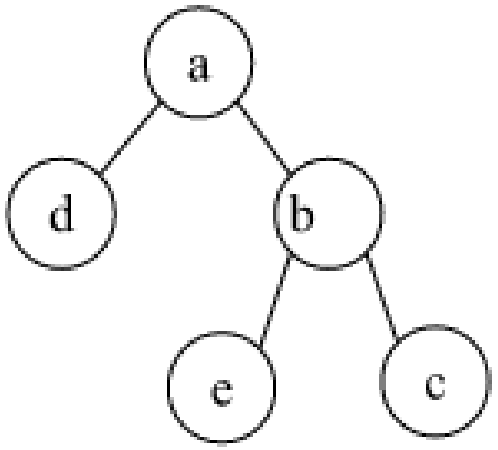
1



3

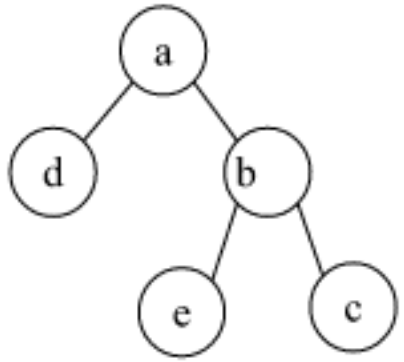
DFS – Exercise 1

- Let $G = (V, E)$ be an undirected graph where $V = \{a, b, c, d, e\}$
- $DFS(a)$ produces the following tree:

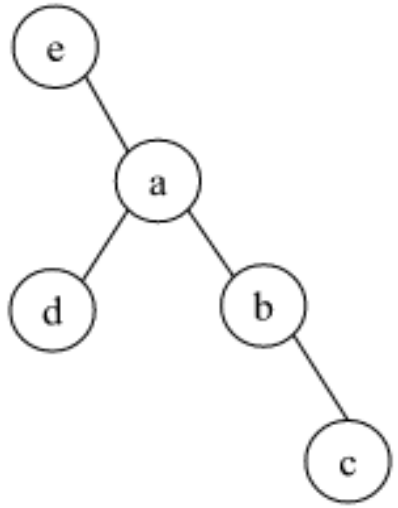


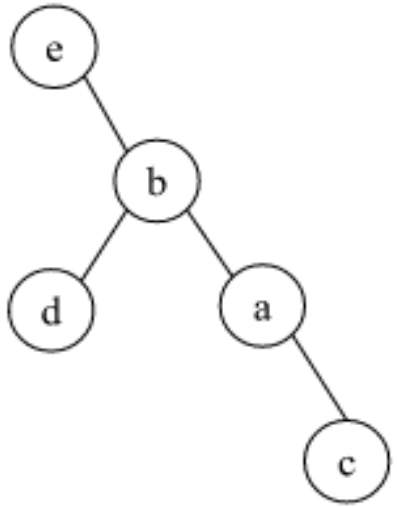
DFS – Exercise 1

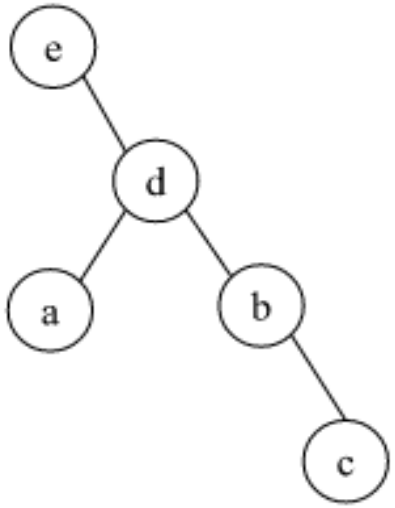
$DFS(a)$



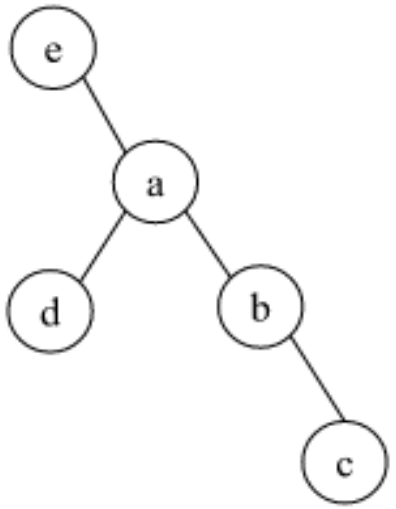
- Which of the following trees can $DFS(e)$ output?

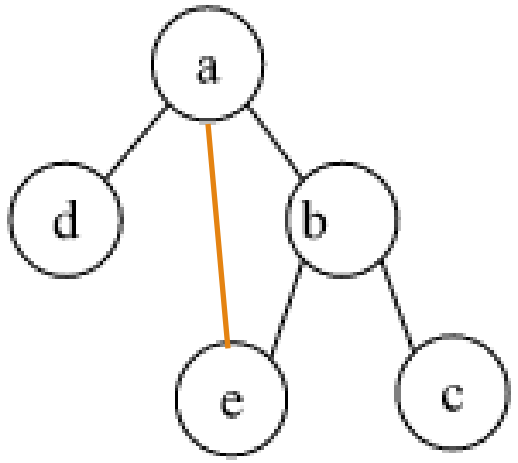






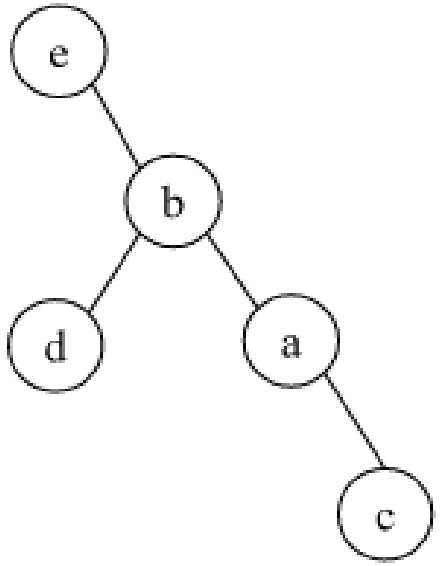
DFS – Solution 1

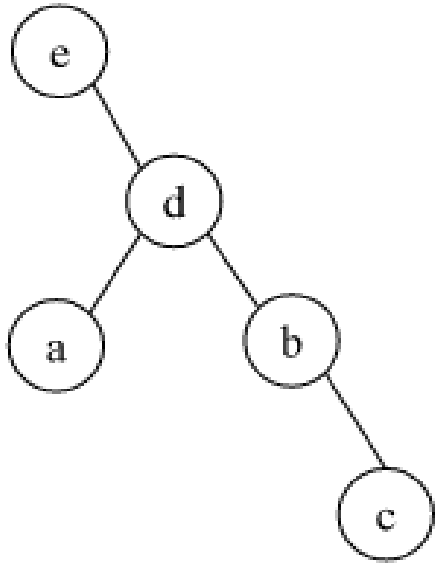




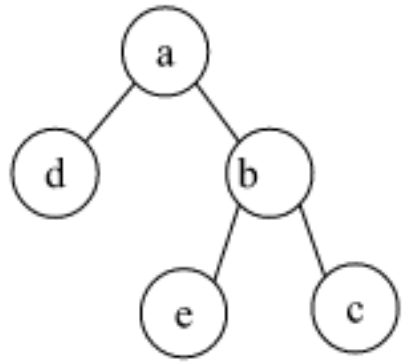
Can be derived if G consists of the given tree with the edge (ae)

DFS – Solution 1





$DFS(a)$



Both of the trees include the edge (bd). If this edge were in G, then d should have been marked by b and not by a in $DFS(a)$

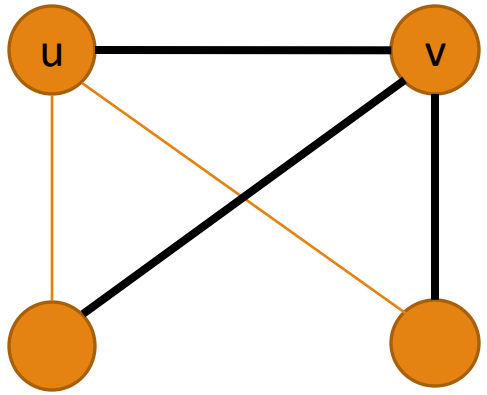
DFS – Exercise 2

- Let $G = (V, E)$ be an undirected graph.
  In the DFS tree of some run $DFS(u)$, a vertex $v \neq u$ has x children.
- True/False: In every run of $DFS(v)$, v has at least x children.

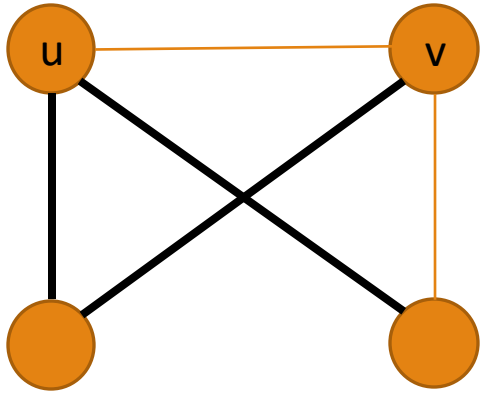
DFS – Solution 2

- True/False: In every run of $DFS(v)$, v has at least x children.

- False!



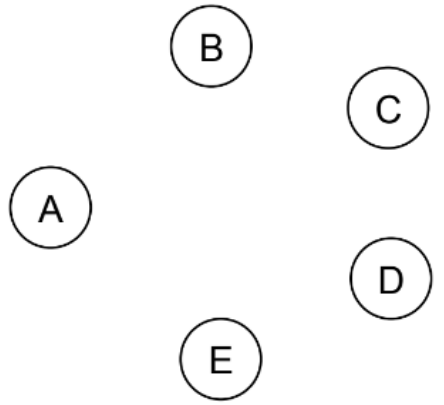
When starting from u -
v has 2 children



When starting from v -
v has a single child

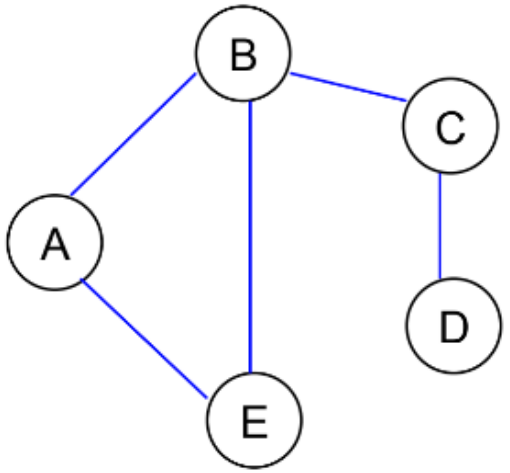
DFS – Exercise 3

- Complete the graph to obtain an undirected graph G that satisfies the following conditions:
  - For every run of $DFS(B)$, in the resulting spanning tree, $deg(B) = 2$.
  - G has a Hamilton path and not a Hamilton cycle.
  - There is a vertex v with $\deg(v) > 2$.

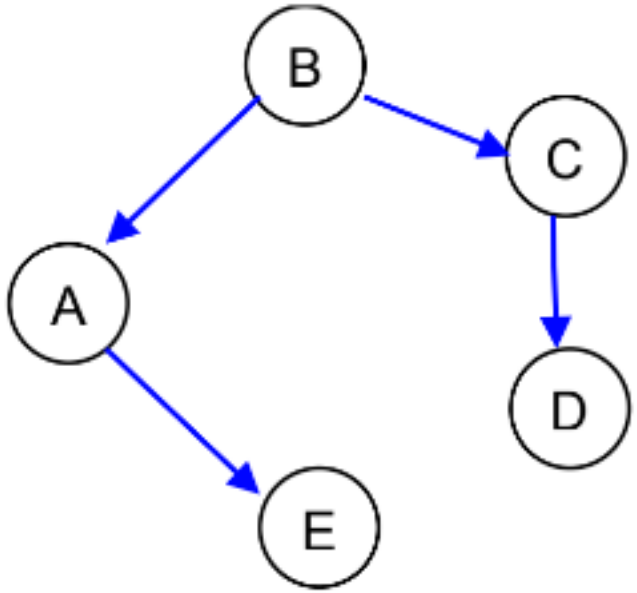


DFS – Solution 3

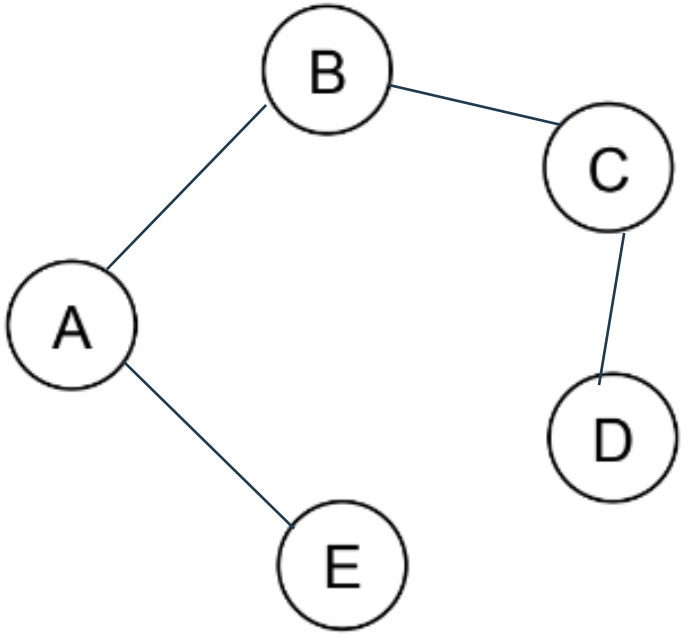
- Complete the graph to obtain an undirected graph G that satisfies the following conditions:
  - For every run of $DFS(B)$, in the resulting spanning tree $deg(B) = 2$.
  - G has a Hamilton path and not a Hamilton cycle.
  - There is a vertex v with $\deg(v) > 2$.



DFS – Solution 3



Possible $DFS(B)$ tree

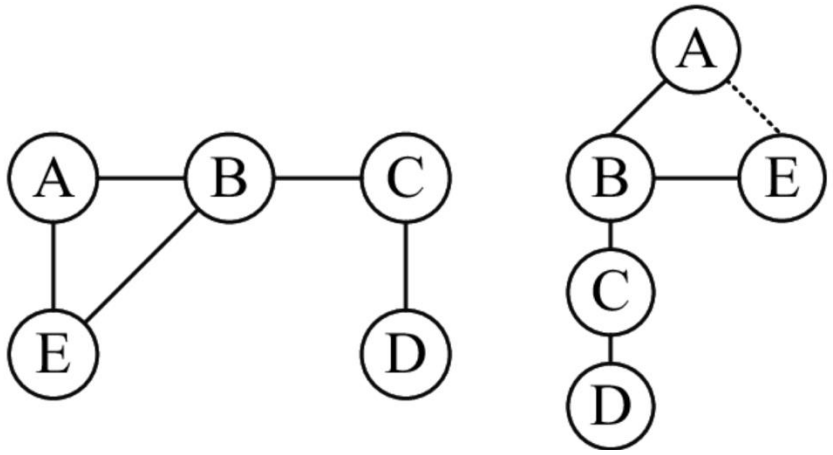


Hamilton Path

Spanning Tree

When a spanning tree is produced by DFS, every edge in an undirected graph G is either a tree-edge or a back-edge.

- Tree edges – edges which belong to the spanning tree.
- Back edges – edges which point from a node to one of its already marked neighbors.



Exercise 4 – True/False

- If there are no back-edges then there are no cycles in G.
- Solution:
  - True. If there are no back-edges then all of the edges in G are tree-edges, therefore G is a tree and therefore it does not include cycles.

Exercise 5

- Let $G = (V, E)$ be an undirected graph with a spanning tree $T = (V, E')$. How many back edges are there in G?

- Solution:
  - There are $|E| - |E'|$ back edges
  - T has $|\mathrm{V}| - 1$ edges
  - Thus, there are $|\mathrm{E}| - |V| + 1$ back edges in G.

Exercise 6 – True/False

- Let G be an undirected graph. Let the vertices $a, b, s \in V$ be such that **every** path between a and b in G goes through s.
- True/False: If we perform $DFS(s)$ then $deg(s) > 1$ in the resulting spanning tree.

Solution 6

- Solution: True
  - The algorithm must visit both a and b. Assume towards contradiction that $\deg(s) = 1$ in $T = DFS(s)$, then both a and b must appear in the subtree T' that results from deleting s from T.
  - Since T' is a tree, a, b are connected in it, therefore there is a path p between a, b in T'.
  - Since s is not in T', we have a contradiction.

Exercise 7

- Let G be an undirected, connected graph. The tree produced by performing $DFS(a)$ on G is given in the leftmost figure. Which of the following trees can be produced by $DFS(b)$?

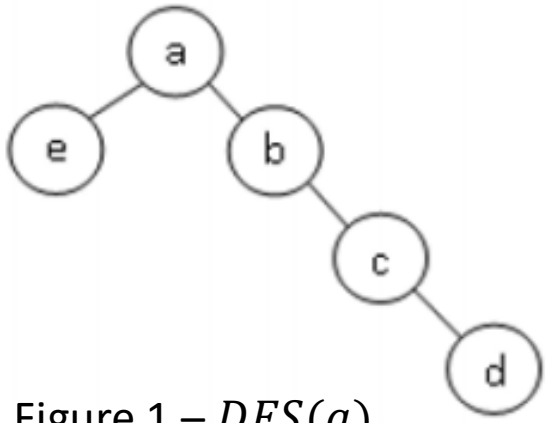


Figure 1 – $DFS(a)$

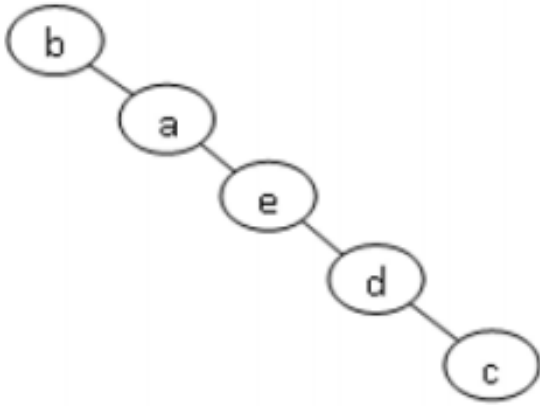


Figure 2

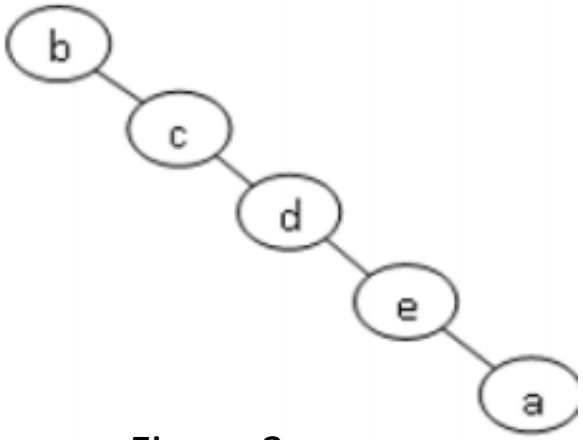


Figure 3

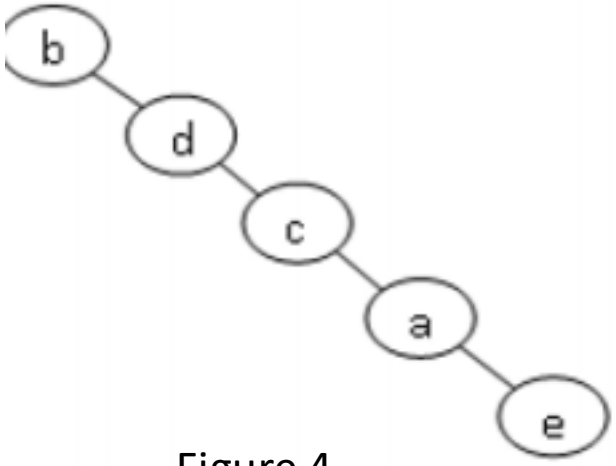


Figure 4

Solution 7

Cannot be $DFS(b)$. Assume that it is, then there is an edge (e, d) in G. It means that $DFS(a)$ should look like: $a \to b \to c \to d \to e$, contradiction.

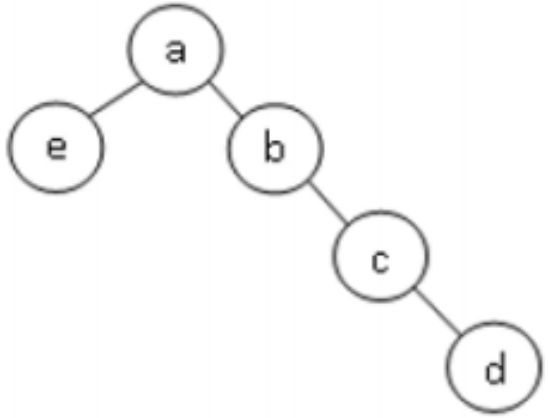


Figure 1 – $DFS(a)$

Cannot be $DFS(b)$. Same argument.

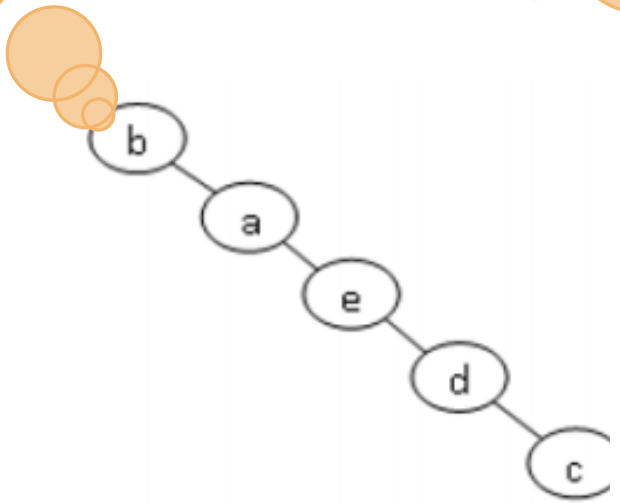


Figure 2

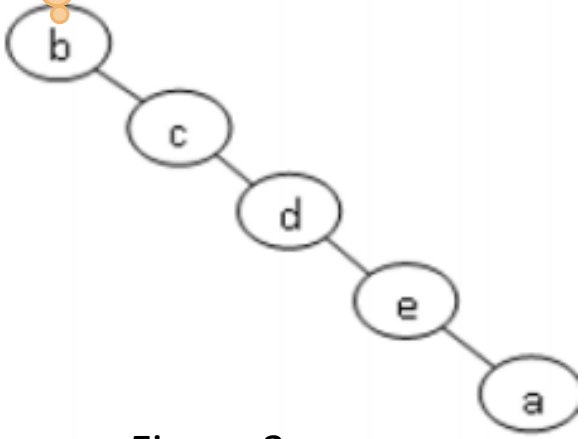
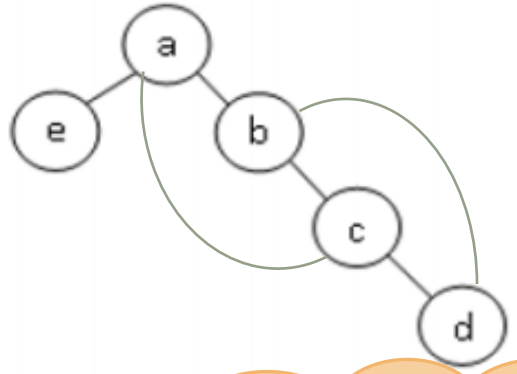


Figure 3

Can be $DFS(b)$. Consider the graph in figure 1 with the additional edges (a, c) and (b, d).



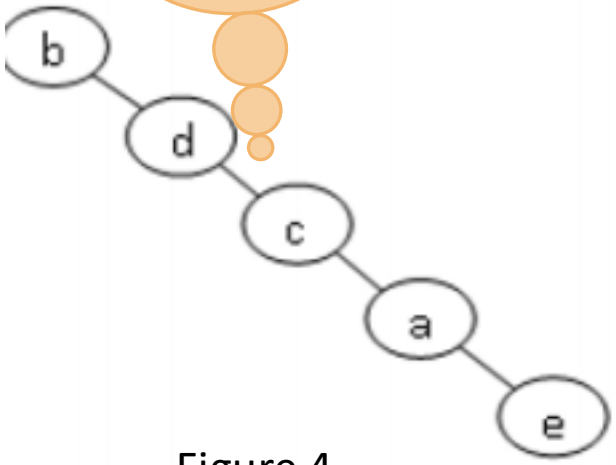


Figure 4